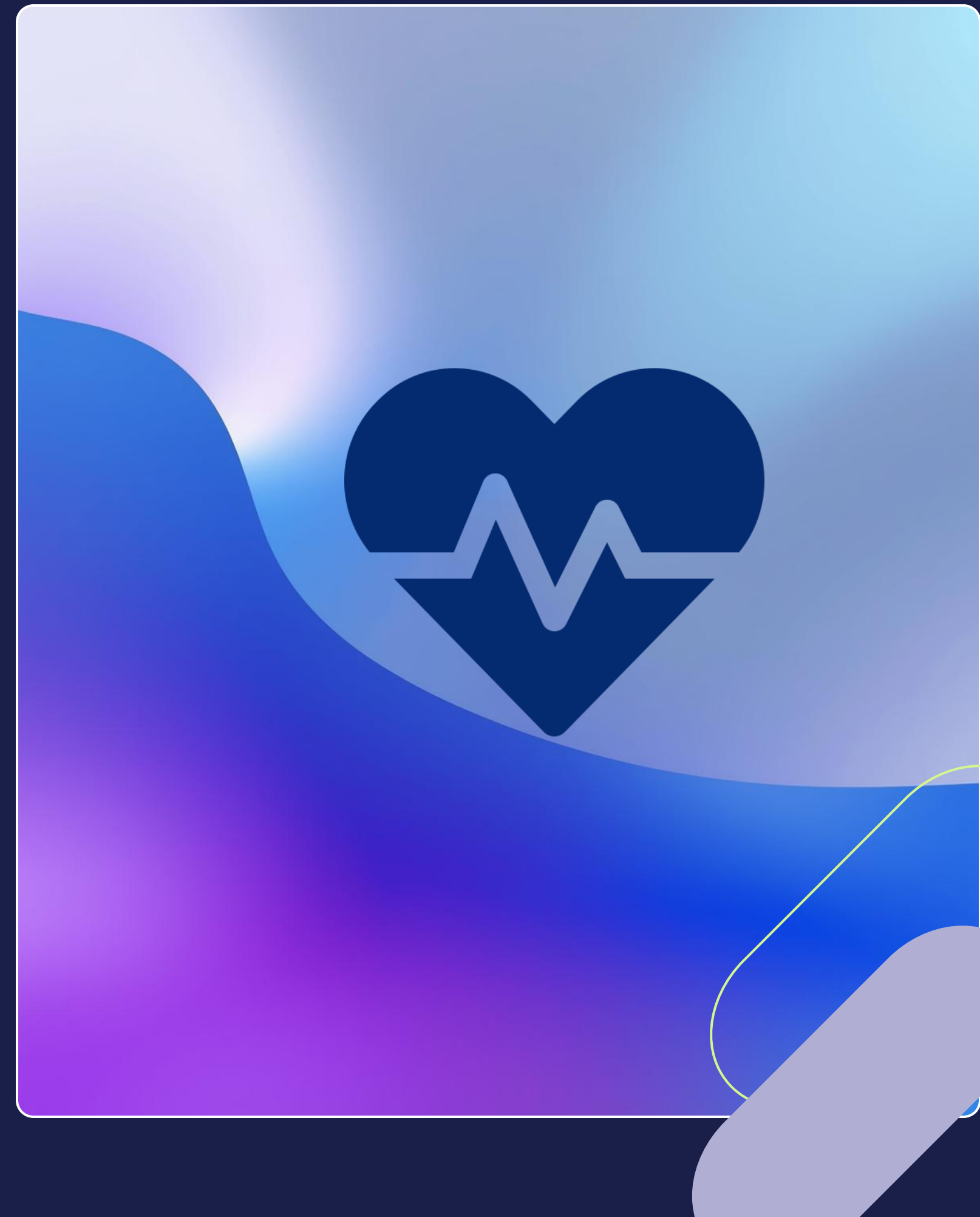


MedTrack

Hospital Operations & Patient Analytics Dashboard

FINAL PROJECT PRESENTATION

By Komal Harshita



Problem Statement

Hospital operational data needs structured analytics.

Structured hospital data must become an interactive analytical view that supports operational decision-making.

1

Hospital Admissions

Tracking admission volumes and trends across the hospital and its departments.

2

Patient Movement

Understanding transfers, peak hours and patient flow patterns.

3

Department Performance

Comparing stay duration, readmissions and efficiency across departments.

4

Resource Utilization

Monitoring beds, staff and equipment to support operational efficiency.

MedTrack: Proposed Solution

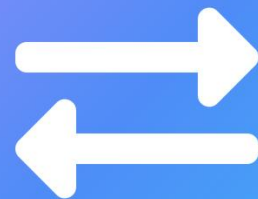
A multi-dashboard hospital analytics solution that transforms cleaned operational and patient data into interactive Tableau dashboards.



DASHBOARD 1

Hospital Performance

Admissions, occupancy, stay duration and efficiency at a glance.



DASHBOARD 2

Patient Flow

Movements, transfers, shifts and peak-hour patterns.



DASHBOARD 3

Department Performance

Compare beds, stay duration, readmissions and efficiency.



DASHBOARD 4

Resource Utilization

Beds, clinical staff and equipment usage and status.

Dataset Architecture



COMMON ANALYTICAL KEYS

hospital_id

department_id

patient_id

admission_id

date

Keys apply where present in each dataset

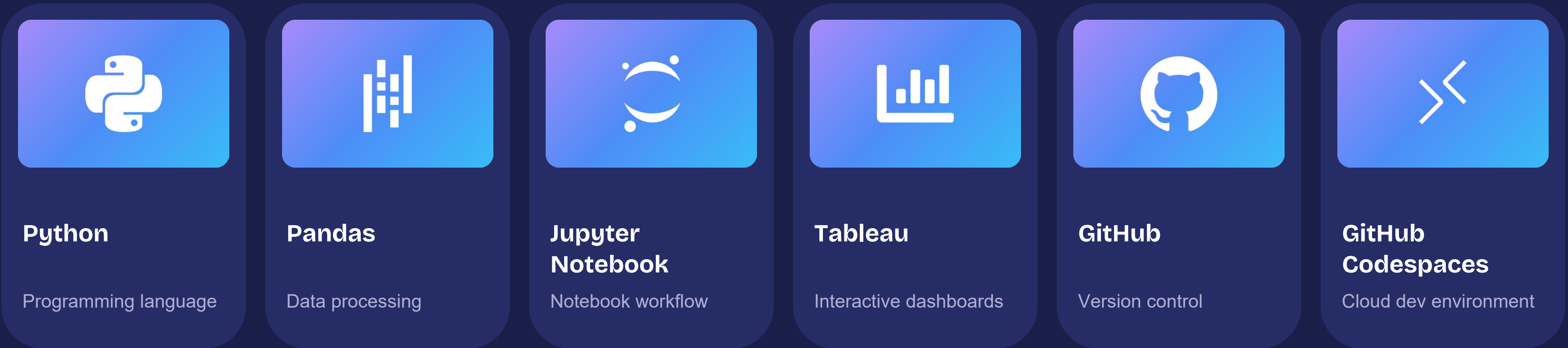
Different grains: no blind joins

- Each dataset is analysed at its own grain
- Physical joins can duplicate metrics
- Related via common keys / Tableau relationships

Data Pipeline & Technology Stack



Technology Stack



KPI Framework: Six Core KPIs

1

Total Admissions

Volume of patient admissions

2

Occupancy Rate

Share of beds that are occupied

3

Average Length of Stay

Typical duration of a patient stay

4

Readmission Rate

Patients returning after discharge

5

Bed Utilization Rate

How effectively beds are used

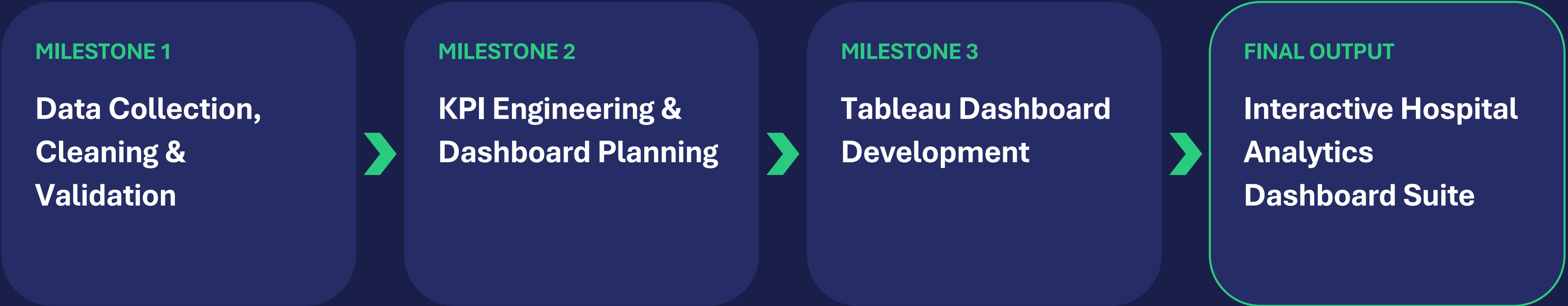
6

Department Efficiency Score

Department-level efficiency measure

Definitions and calculations follow the project's KPI documentation.

Project Milestones



PROJECT OUTPUTS



Hospital Overview

DASHBOARD 1 OF 4

The high-level hospital performance view.



KPI CARDS

- Total Admissions
- Occupancy Rate
- Average Length of Stay
- Readmission Rate
- Bed Utilization Rate
- Department Efficiency

VIEWS

- Admissions Over Time
- Admissions by Department
- Admissions vs Discharges
- Operational Snapshot

KPI summary cards

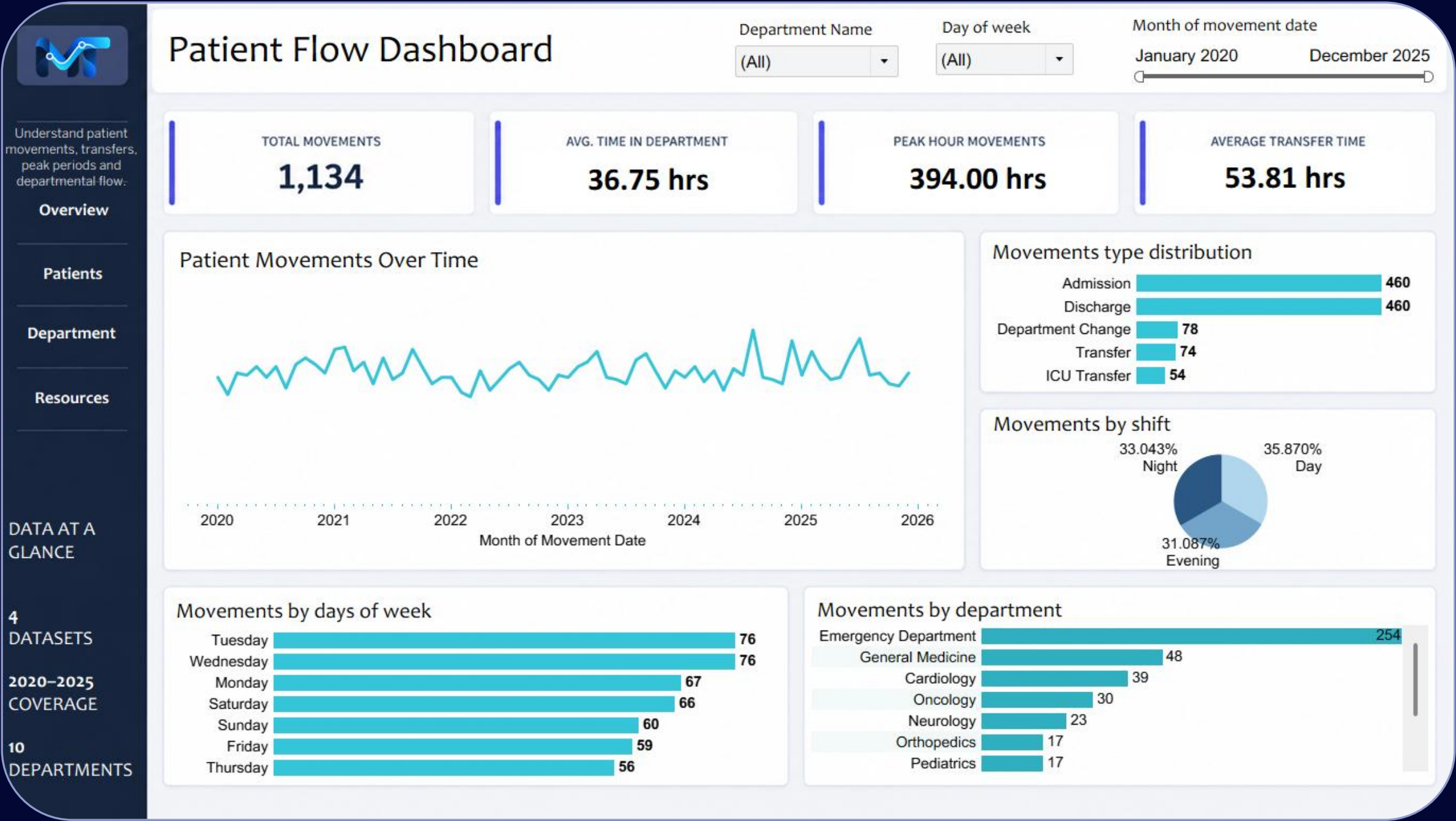
Admissions trend

Department split

Operational snapshot

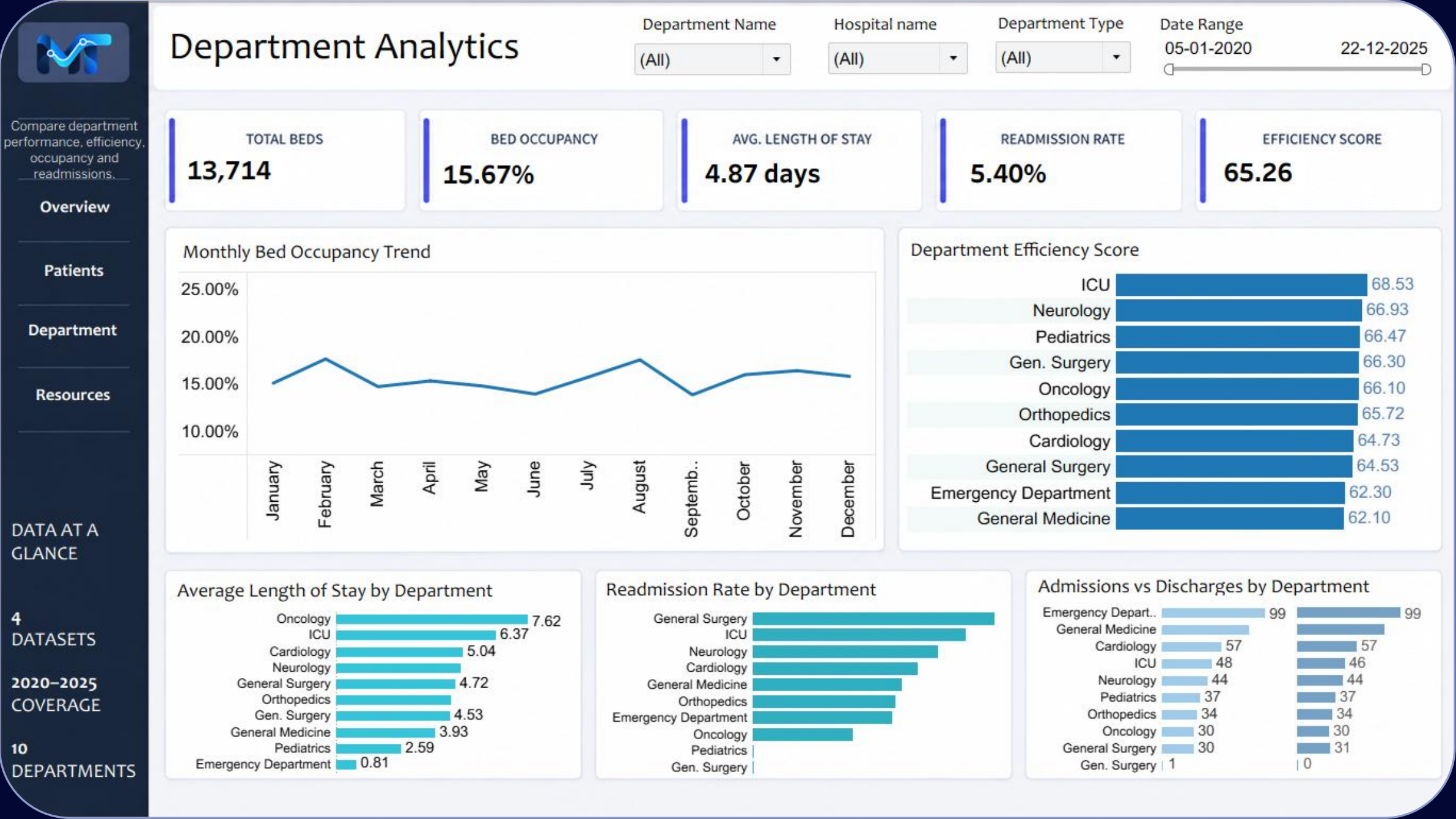
Patient Flow

Understanding patient movement and flow patterns.



Department Analytics

Comparing department-level performance.



KPI CARDS

- Total Beds
- Bed Occupancy
- Average Length of Stay
- Readmission Rate
- Efficiency Score

VIEWS

- Monthly Bed Occupancy Trend
- Department Efficiency Score
- Avg Length of Stay by Department
- Readmission Rate by Department
- Admissions vs Discharges by Dept

Occupancy trend

Efficiency by department

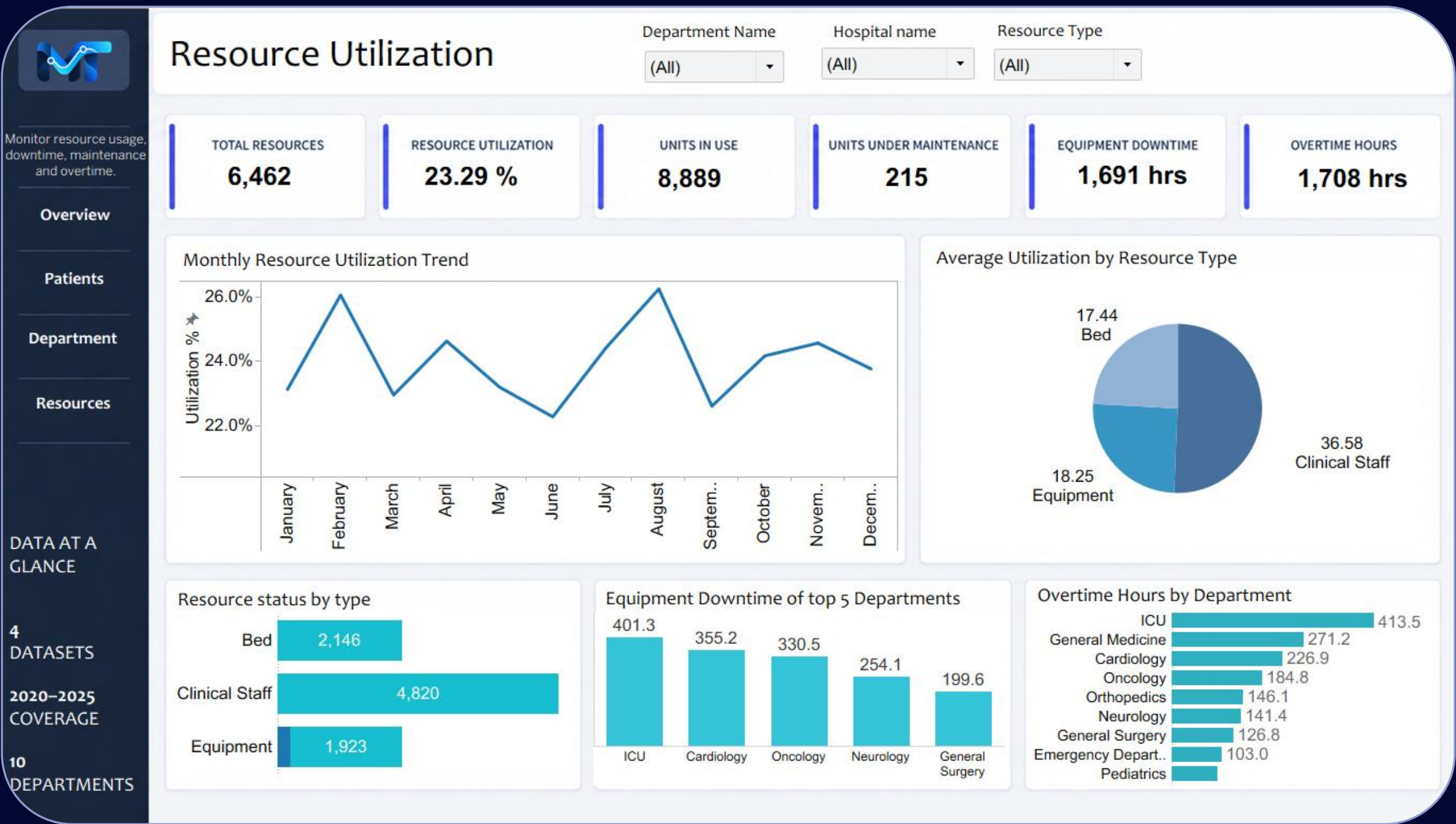
Stay & readmissions

Admissions vs discharges

Resource Utilization

DASHBOARD 4 OF 4

The operational resource management view.



KPI CARDS

- Total Resources
- Resource Utilization
- Units in Use
- Units Under Maintenance
- Equipment Downtime
- Overtime Hours

VIEWS

- Monthly Utilization Trend
- Avg Utilization by Resource Type
- Resource Status by Type
- Equipment Downtime by Dept
- Overtime Hours by Dept

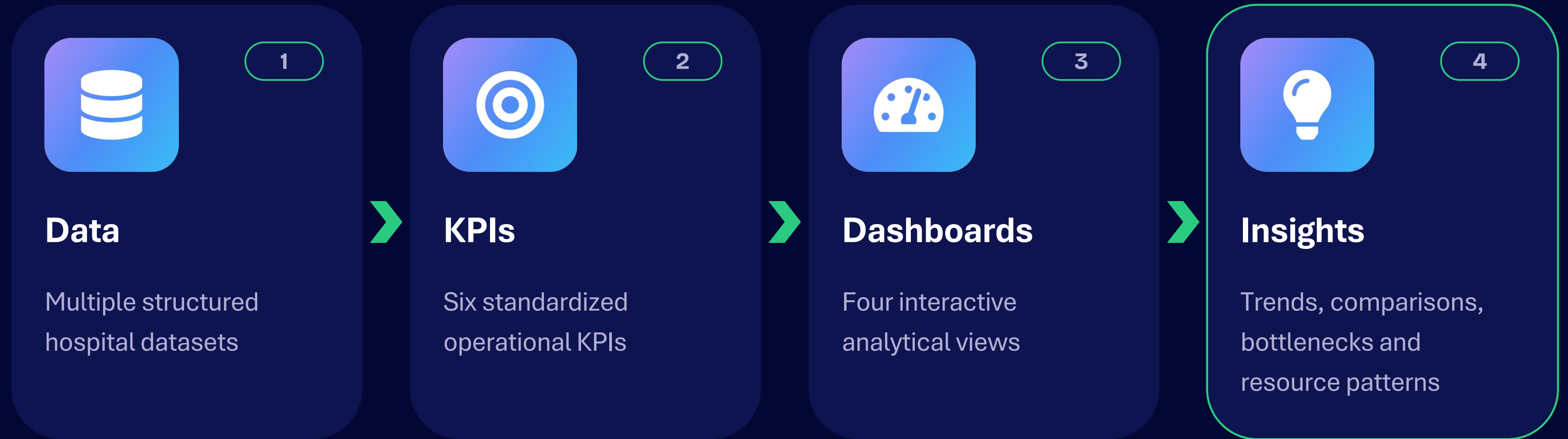
Beds

Clinical Staff

Equipment

Downtime & overtime

From Data → KPIs → Decisions



Decision support, not decision-making. MedTrack helps users explore operational patterns; it does not make clinical decisions.

Challenges & Solutions

FUTURE SCOPE

Not part of the current build

- Advanced forecasting
- Automated anomaly detection
- Granular operational alerts
- Hospital-level benchmarking
- Automated data refresh pipelines
- Advanced predictive analytics

1

Different Dataset Grains

SOLUTION

Separate analytical datasets and Tableau relationships instead of blind joins.

2

Data & KPI Consistency

SOLUTION

Standardized IDs, categories and dates; six mandatory KPIs defined and validated.

3

Dashboard Complexity

SOLUTION

Analysis split into four focused dashboards.

4

Usability

SOLUTION

Filters, navigation and dashboard-level analytical views.

MedTrack

Hospital Operations & Patient Analytics
Dashboard

“From hospital data to actionable operational insight.”

Thank You

Q & A

